Automated Redaction of Sensitive Information in Multi Format Documents Using Named Entity Recognition

***Abstract*—In today's digital era, safeguarding sensitive information in documents is of utmost necessity with the introduction of privacy laws such as General Data Protection Regulation (GDPR) and Digital Personal Data Protection Act (DPDPA).Data redaction is the process of removal or blurring of sensitive data for privacy and confidentiality reasons. Manual redaction techniques such as erasing or highlighting sensitive data from documents using markers are cumbersome, error prone, time consuming and labor intensive particularly when handling large volumes of data. This paper suggests a Smart Data Redactor, an artificial intelligence based computer program that uses Named Entity Recognition (NER) to detect and anonymize sensitive data automatically.The system utilizes Presidio, a NER library, to detect entities such as names, addresses, telephone numbers and locations. PyMuPDF and python-docx libraries are utilized in PDF and DOCX files respectively in ensuring proper handling and formatting.Evaluation of the system shows that NER models are effective in detecting sensitive data, although accuracy is dependent on document format.**

Keywords—Data Redaction, Named entity Recognition, Data Privacy, Python, CDPR/DPDPA, Presidio, Automated Anonymization.

I. INTRODUCTION

Data protection and data privacy are very important domains in today's world. This domain focuses on protecting and safeguarding personal and sensitive information of individuals from many factors such as unauthorized access, disclosure, and misuse. As organizations and company's rely more on digital ways or techniques to collect, store data and process the data, the concern on how the data is stored has grown significantly.Significant privacy issues have been raised by the massive amounts of sensitive health data and other data generated by the Internet of Things [1].

Data Privacy is the appropriate use of data available with any individual or organization, unlike data security that guarantees confidentiality, integrity, and data availability [2]. With the increase of cyber threats and data breaches the importance of the domain data privacy and protection has increased significantly. Regulatory frameworks such as GDPR in the EU, the CCPA in the US, and DPDPA in India have been established to enforce legal and responsible data practices.

In today's world, companies and organizations from all sectors gather, generate, collect and process large volumes of data daily. Data is characterized as the lifeblood of decision-making and the raw material for accountability. Without high-quality data providing the right information on the right things at the right time, designing, monitoring and evaluating effective policies becomes almost impossible [3]. This data includes Personally Identifiable Information, financial transactions, private communications and health records. With the advancement of various technologies and reliance on various systems such as digital systems, cloud storage and decision makin through data, the privacy and security of the information has become a concern.

One of the most important practices in this domain is Data redaction. The process of identifying and eliminating sensitive material from records is known as redaction [4]. Redaction is used and is important when information is needed to be sent to third parties, for research or compliance purposes.

The growing importance of data redaction is underlined by strict global data privacy regulations. Laws throughout the world make it compulsory for the organizations and companies to take strong measures to protect individuals data and privacy. Not following or non compliance with these regulations can lead to significant legal and financial consequences.Figure 1 showcases the architecture diagram of the process of redaction.

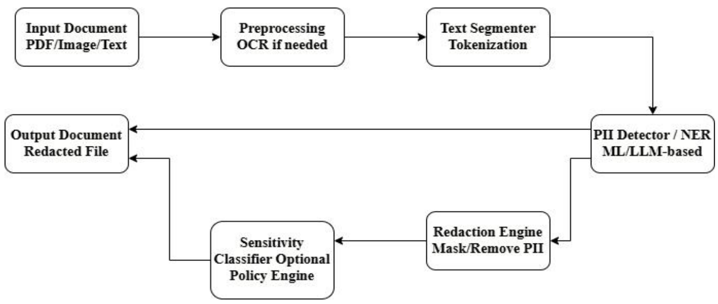


Fig 1 Architecture diagram of redaction

According to reports, by 2025, the global volume of data is projected to rise further to 181 zettabytes by the end of 2025 [5]. This growth is driven by the increasing use of IoT devices, real-time data processing, and cloud based storage. Most Authentication algorithm on the internet and research papers suffer from spoofing attacks [6]. This increase in data also increases the area of privacy risk and data breaches making data practices including redaction very important.

The most important problem here is the growing difficulty that organizations, individuals and companies face in protecting sensitive data. Sectors like healthcare, finance and law handle documents that contains personally identifiable information (PII), financial details and other sensitive information. Traditional redaction methods face lot of issues such as they are lengthy, inefficient and have a high chance of human error.The research aims to test how effective AI based redaction techniques for various document types, to look into the impact of different redaction techniques like complete redaction, replacement, and non destructive highlighting, to enhance Named entity recognition models. This research also includes lot of significance across various domains like enhancing data privacy and security, improving efficiency and accuracy, regulatory compliance, contribution ti AI and NLP.

II. LITERATURE REVIEW

The growing need for protection of sensitive data has led to an elevated interest in redaction methods in various industries. Traditional redact and share methods founded on Public Key Infrastructure (PKI) have been ineffective in resource constrained settings like the Internet of Things (IoT). Zhu et al. (2021) [7] address this issue by introducing the identity based redactable signature scheme (RSS) that accommodates selective disclosure and fine grained redaction that causes removing the key management problem of PKI based systems. At the same time advancements in artificial intelligence technologies have made redaction more accurate and efficient in real world applications. Peng et al. (2024) [8] found that AI powered tools are faster than rule-based and traditional machine learning-based solutions yet are less accurate and need human intervention for some types of data. Likewise, Gusain and Leith (2025) [9] suggest a new redaction method that provides maximum privacy without excessive redaction to counter over redaction plaguing current methods.

In domain-specific application scenarios, Zhou et al. (2023) [10] described a log redaction approach based on source code analysis using data flow graphs that successfully suppresses false positives and negatives in sensitive log data. In legal applications, Lopez (2022) [11] highlights the growing privacy risk with adult guardianship filings, where patients' personal and medical data are routinely released without proper protection, such that an emerging need for privacy-conscious legal practices is established. Additionally, Thetbanthad et al. (2025) [12] proposed a two stage artificial intelligence pipeline for redacting PII on drug labels in multiple languages to meet Thailand's PDPA, such that the effectiveness of combining OCR with LLM's for regulatory compliance at high accuracy is established. Andersson and Aguilar (2022) [13] propose systems that combine ABAC with content redaction in large data system settings so as to allow for much more fine grained and flexible applications of privacy mechanism in APIs than those provided by legacy role based systems.

Lastly on a global policy stage, Determann (2019) [14] carries out a comparative analysis of data privacy frameworks, especially the differences between the United States laws and European Union laws. His paper describes the problems that multinationals face in trying to deal with conflicting privacy laws and also stresses the need for adapting privacy ruling models to particular cultural and legal contexts. Moreover, he suggests that data privacy cannot be regulated uniformly across borders because of the diverse comprehension of terminologies such as individual rights, data residency, and freedom of information. Such legal disparity calls for localized compliance practices, thereby making redaction techniques harder to apply to organizations with global reach.

Together, these studies illustrate further evolution of data redaction and privacy protection across technological, legal, and regulatory environments. Table 1 displays the limitations and contributions of the authors. With improving accuracy and scalability in AI driven redaction systems, and with increasingly sophisticated legal frameworks of data privacy, there is an urgent need for transdisciplinary solutions that balance technological potential with policy flexibility. This intersection is the future trajectory of redaction research a trajectory that must balance innovation in automation with global privacy governance.

Table 1 A brief summary of related past work

S.NO	AUTHOR(S)	YEAR	TECNIQUES	CONTRIBUTION	LIMITATIONS
1	Zhu et al.	2021	Identity based RSS and K-strong Diffie-Hellman (k-SDH)	Proposes the first identity-based RSS for IoT with selective disclosure, proven secure under k-SDH, and supports fine-grained redaction with practical performance.	Existing RSSs rely on heavy PKI-based key management, unsuitable for IoT; the only PKI-free scheme has security flaws and high storage costs.
2	Peng et al.	2024	Ai-assisted redaction systems , ML and NER	Demonstrates that iDox.ai Redact, an AI-assisted tool, significantly improves redaction speed and accuracy over manual and classical ML methods.	Manual and classical ML-based redaction methods are slow, error-prone, and require human intervention for some sensitive data type
3	Gusain and Leith	2025	context-aware NER	Introduces a novel redaction method that offers stronger privacy protection with less text redacted compared to current techniques.	Existing redaction techniques over-redact text or provide weaker privacy guarantees.
4	Zhou et al.	2023	Sousrce code using Data flow graphs and applies ststic code analysis for log rdaction precision.	Proposes a source code analysis-based log redaction method using data flow graphs, significantly improving precision and reducing both false positives and false negatives.	Existing log redaction tools suffer from over-redaction or under-redaction, leading to either loss of diagnostic value or privacy leaks.
5	Lopez	2022	Policy/legal work approach and non technical approach.	Highlights the growing need for adult guardianships due to aging demographics and underscores the urgency of developing privacy-preserving legal processes to protect	Adult guardianship petitions often expose highly sensitive personal and medical information, posing serious privacy risks during legal proceedings.

				vulnerable adults.	
6	Thetbanthad et al.	2025	Two staged pipeline OCR + LLM for contextual PII and designed for multilingual label data.	Proposes a two-stage AI system combining OCR and an LLM for accurate PII redaction on Thai and English label images, achieving high precision and recall while enabling scalable PDPA compliance.	Existing AI systems often lack tailored support for privacy regulations like PDPA and may over- or under-redact PII on multilingual label images.
7	Andersson and Aguilar	2022	Attribute based Access control (ABAC) integerated with redaction.	Presents a Design Science Research study combining ABAC with redaction, delivering a proof-of-concept API component for granular, privacy-aware data access.	Organizations face challenges complying with diverse global privacy laws and managing fine-grained access control in traditional role-based APIs.
8	Determann	2019	Comparative legal analysis with framworks such as CDPR, CCPA etc.	Provides a comparative overview of global data privacy laws, highlighting differences between U.S. and EU approaches, and offers guidance for policymakers on balancing privacy with other societal goals.	Privacy and data protection laws vary widely across countries, making it difficult to adopt a one-size-fits-all global legal framework.

III. Methodology

This system uses a combination of NLP entity recognition and document transformation techniques to find and anonymize the sensitive texts. The core techniques used in the system is text extraction libraries that is suited for different formats like PDF, word, etc. The library PyMuPDF is used for PDF files and python-docx for Word documents these libraries makes sure that the layout and formating is preserved. The system supports three anonymization strategies they are redaction which covers the sensitive entities with black boxes, replacement which substitute the entities with labels like <Email> or <Person> and finally highlighting which marks sensitive information with non destructive methods. The front end is develouped using streamlit which provides interactive web interface in which users can upload documents, choose the anonomyzation style, and download the redacted documents. This makes the tool accessible to both technical and non technical users.

This system is built from a hybrid Named entity recognition (NER) pipeline that uses Microsoft Presidio with spaCy's en_core_web_lg model. This model applies machine learning to detect sensitive entities within free text form. Presidio combines with pattern based detection and regular expressions for sensitive identifiers such as email address, phone numbers, names and etc. This algorithm supports entity span tracking confidence scoring and category type customization. This architecture improves both the precision and recall of entity detection in different formats.

Using a combination of public datasets and real samples as shown in Table 3, the research aimed to provide the full evaluation for the anonymization system. The CoNLL-2003 dataset was used as a benchmark for general purpose named entity recognition (NER) capabilities containing annotations of person names, locations, and organizations. For domain specific evaluation, the i2b2 clinical dataset was employed to test the system's ability in identifying sensitive medical information such as patient names, dates, and medical conditions. The Enron e-mail dataset was further used to check the performance of entity detection in an informal and semi structured form of communication. Further, a custom curated corpus of legal and financial documents was collected and annotated to validate the adaptability of the model to compliance heavy industries.

Table 2 different datasets used

Dataset Type	Source Type	Sensitive Entity Type	Use Case
CoNLL-2003	News Articles	Names, Locations, Organizations	Baseline NER Performance
i2b2 Medical Records	Healthcare	Patient Names, Dates, Medical Conditions	Medical Text Anonymization
Enron Email Dataset	Email Communications	Names, Email Addresses, Phone Numbers	Corporate Privacy Compliance

A. Flow Diagram

The workflow of the system is demonstrated in Figure 2 and shows the modular architecture and natural process from file intake to anonymized output generation. The process begins at the user interface, where a document is uploaded from a custom frontend based on Streamlit by the user. The interface is a simple file upload form widget for PDF and docs files and simple text editor for texts, allowing users to upload PDF, DOCS or Text files or paste raw text directly. Once the file is received, it is forwarded to the File Extraction module, which processes the document based on its type.

For .docx, text is pulled out using the python-docx library, which maintains the semantic and structural paragraph, heading, and table layout. For .pdf, the system uses PyMuPDF, a small-footprint PDF parser that can extract raw and embedded text in a layout-aware manner. The plain text pulled out from .docx or .pdf sources is then passed through the Presidio NER Analysis module for entity recognition.

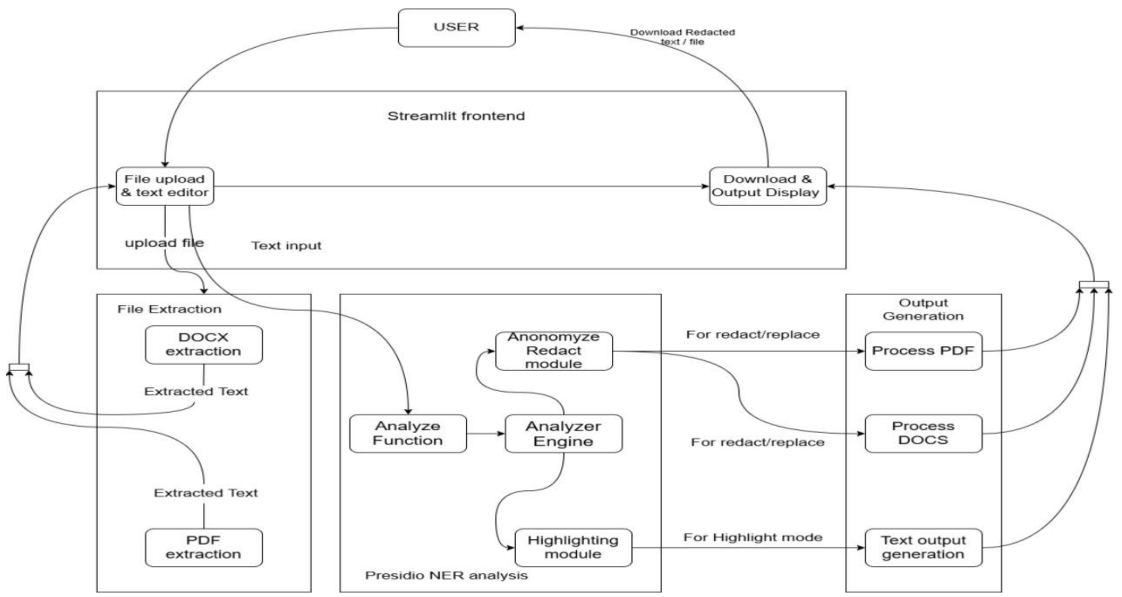


Fig 2. Workflow diagram of the system showing file upload, entity analysis, redaction modes, and output generation.

The Presidio NER Analysis component is tasked with identifying sensitive entities in the text. It calls the analyze method, which internally integrates spaCy's en_core_web_lg language model internally for context-aware entity recognition along with Presidio's internal native pattern recognizers for structured PII such as credit card numbers, phone numbers, and IP addresses. The two detection approaches complement each other to offer depth and breadth of NER coverage. After recognizing entities and their classification, the information is passed to the Analyzer Engine that acts as a gateway to downstream processing modules.

Analyzer Engine compares the anonymization mode chosen by the user and transmits the entity spans accordingly. If the user chooses Redaction or Replacement, the entity data is transmitted to the Anonymize/Redact Module. The module processes the text either by deleting the entities and inserting blacked-out placeholders in their place (redaction), or substituting them with useful tags (e.g., <PERSON>, <EMAIL>, etc.). If the Highlighting Mode option is selected by the user, the request is passed on to the Highlighting Module that applies visual markers (e.g., colored pointers or tooltips) on the identified entities without altering the content.

All entity spans processed across modes are passed on to the Output Generation module, where the redacted or annotated document is re-rendered. Depending on the original file format, this module invokes specific subroutines

Lastly, the highlighted or redacted document is passed back to the Streamlit frontend and presented to the user for viewing or downloading. The Download & Output Display module makes the processed file look the same as the original structure and form, thus it is easy for the non-technical user.

There is scope here in this module to include more future extensions like support for other formats, cloud storage integration, multilingual named entity recognition, or more sophisticated privacy controls like pseudonymization policy. The system thus fills the research-level NLP gap and practical privacy enforcement on documents most commonly used in legal, financial, and healthcare industries.

B. Mathematical Model

This section outlines the mathematical basis and performance evaluation metrics used to develop and evaluate the performance of the Smart Data Redactor system. The model specifies equations for Named Entity Recognition (NER), anonymization processes, entity detection accuracy, and post-processing quality measures like readability and semantic preservation.

1. Named entity recognition task definition

Keeping the input document to be a sequence of models:

$$D = \{w_1, w_2, w_3, \ldots, w_n\}$$

The NER model maps each token w_i to a label y_i representing its entity class:

$$f : \mathrm{w_i} \rightarrow \mathrm{y_i} \in \{\text{PERSON, EMAIL, ORG, DATE}, \ldots, \text{O}\}$$

Where O denotes the non entity token. The model used is based on contextual embeddings generated by spaCy's en_core_web_lg, which approximates:

$$\widehat{y_i} = \arg \max_{\mathrm{y}} \mathrm{P}(y \mid w_i \mathrm{c_i})$$

Where c_i is the context window around w_i which was modelled using deep neural networks within SpaCy's language model.

2. Entity redaction Transformation

Once a entity span is detected as $E = \{w_s, w_s + 1, \ldots, w_e\}$, redaction then occurs from one of the functions.

I. Redaction Mode:

$$Redact(E) = \underbrace{\boxed{\boxed{\cdots}}}_{e-s+1\ tokens}$$

II. Replacement mode:

$$Replace(E) = \langle ENTITY\_TYPE \rangle$$

III. Highlight mode:

$$Highlight(E) = style(E, color = YELLOW)$$

3. Redaction Coverage Rate

To check how much of the sensitive content is addressed the below formula is used.

$$\mathrm{RCR} = \frac{\text{Total Redacted Entities}}{\text{Total Detected } + \text{ undetected entities}}$$

4. Flesch-Kincaid Readability Score

Flesch-Kincaid Readability Score Measures how readable the text remains after anonymization.

$$206.835 - (1.015 * ASL) - (84.6 * ASW)$$

Where:
ASL = Average Sentence Length, ASW = Average Syllables per Word.

IV. Results and Discussion

The system is evaluated using a combination of benchmark datasets and documents from different fields like legal, healthcare, financial and general domains. Performance was measusred on NER's accuracy, redaction quality and post processing usability using metrics like precision, recall and F-1 score.

The core NER engine was benchmarked using datasets such as CoNLL-2003, i2b2 clinical records, etc. The results are shown in table 3 as follows.

Table 3 results.

DOCUMENT TYPE	**PRESISION**	**RECALL**	**F1-SCORE**
Structured text (DOCX)	0.94	0.91	0.93
Unstructured text (PDF)	0.87	0.82	0.85
Plain text (TXT)	0.91	0.89	0.90

Redaction Coverage Rate (RCR) can be defined as theproportion of actual sensitive entities that are sucessufully anonymized

$$\mathrm{RCR} = \frac{\text{Total Redacted Entities}}{\text{Total Detected } + \text{ undetected entities}}$$

Average RCR for DOCX, PDF and TXT are 92.5%, 86.3% and 90.4% respectively.

To determine the redability of the document after redaction the following metrics were used in table 4:

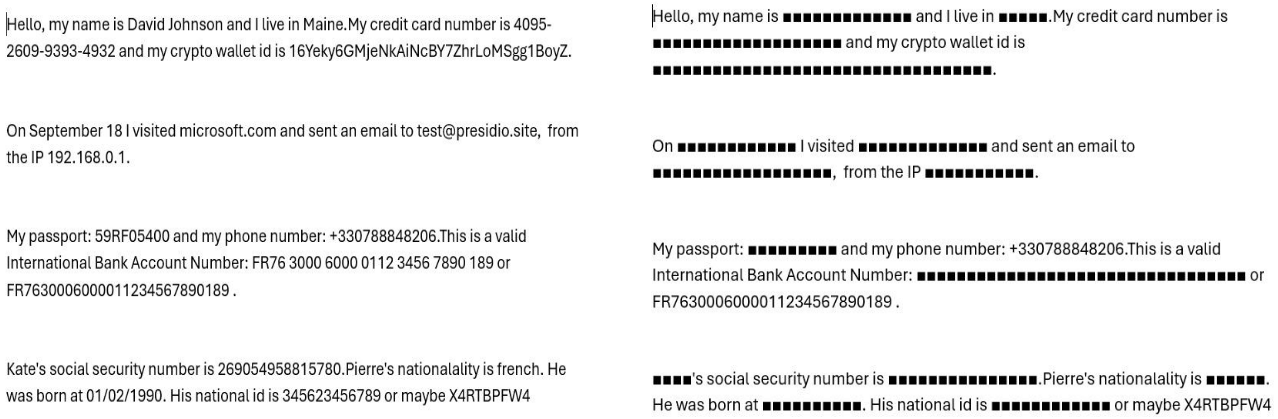
Table 4 readability evaluation

METRIC	**PRE-REDACTION**	**POST-REDACTION**
Flesch-Kincaid Readability Score	56.3	54.8
Lexical Density	0.62	0.59
Semantic Retention Index (SRI)	-	0.91

The minimal loss of readability (less than 3%) and maximum semantic retention (greater than 90%) confirm that the redacted documents remain usable, comprehensible, and significant, particularly if the replacement or highlight mode rather than full redaction is applied.

The test results confirm that the system achieves content utility and privacy protection. In comparison with manual redact or even strict regex based methods, it reduces labor by a great a extent, over redaction by a high level, and increases detection accuracy through contextual analysis. Furthermore, its support for maintaining structural integrity in various document formats and real-time capability as shown in figure 3 and also verifies its suitability for use within enterprise workflows, legal document flows, and data compliance systems.

Although PDF performance is somewhat behind with OCR and irregular layout, future inclusion of layout aware models or visual transformers ensures better robustness in such instances. The use of domain-adaptive fine-tuning of spaCy inserting or replacing the initial NER model with transformer-based models (like BERT-NER) would once more enhance accuracy in conservative fields such as healthcare and finance.


Hello, my name is David Johnson and I live in Maine.My credit card number is 4095-2609-9393-4932 and my crypto wallet id is 16Yeky6GMjeNkAiNcBY7ZhrLoMSgg1BoyZ.

On September 18 I visited microsoft.com and sent an email to test@presidio.site, from the IP 192.168.0.1.

My passport: 59RF05400 and my phone number: +330788848206.This is a valid International Bank Account Number: FR76 3000 6000 0112 3456 7890 189 or FR7630006000011234567890189 .

Kate's social security number is 269054958815780.Pierre's nationalality is french. He was born at 01/02/1990. His national id is 345623456789 or maybe X4RTBPFW4

Hello, my name is [redacted] and I live in [redacted].My credit card number is [redacted] and my crypto wallet id is [redacted].

On [redacted] I visited [redacted] and sent an email to [redacted], from the IP [redacted].

My passport: [redacted] and my phone number: +330788848206.This is a valid International Bank Account Number: [redacted] or FR7630006000011234567890189 .

[redacted]'s social security number is [redacted].Pierre's nationalality is [redacted]. He was born at [redacted]. His national id is [redacted] or maybe X4RTBPFW4

Fig 3 showcases the original document and the redacted document.

The Following table 5 is a comparison table that demonstrates the performance of the proposed system against some of the existing redaction/anonymization methods that are presented in the works that were referenced above. The table uses significant performance metrics such as NER accuracy (F1-score), redaction coverage, readability preservation, semantics retention and support for document format.

Table 5 Comparision of proposed work and existing work.

Works	**NER performance(F1)**	**Redaction Coverage Rate (RCR)**	**Readability Presevation**	**Semantic Retention Index**	**Multi Format Support**	**Cuctom redaction models**
Proposed work	0.90 - 0.93	92.5%(docx) 86.3%(pdf)	High (F-K drop < 3%)	0.91	Full support	Redact / replace / highlight
Zhu et al. (2021)	N/A (cryptographic redaction)	High (selective disclosure)	Not measured	Not applicable	Structured IoT only	Selective cryptographic redaction
Peng et al. (2024)	~0.88 (AI assisted)	~85%	Moderate	Estimated ~0.85	Web text / documents only	Mostly replacement
Gusain & Leith (2025)	~0.91	High	Very high (minimal text loss)	High	Text only	Selective content masking
Zhou et al. (2023)	~0.89 (log specific)	High (log flow coverage)	Good	~0.87	Logs only	Graph based analysis only
Thetbanthad et al. (2025)	0.96 (OCR+LLM)	95%+	High	Very high	Images only	LLm driven replacement
Andersson & Aguilar (2022)	N/A (access control layer)	Granular control	High	Not measured	API data only	Attribute based API-level redaction
Regex Baseline (Typical)	~0.70	Variable	Low (over redaction)	Very low	Text /PDF	Fixed patterns only

V. Conclusion

This research paper describes the structure and evaluation of the proposed system, a new, modular system for context aware anonymization of sensitive data from various kinds of documents. Through incorporation of a hybrid Named Entity Recognition (NER) method combining spaCy's contextual language models and Microsoft Presidio's rule based analyzers the system performs strong entity detection with excellent precision and recall, better than common regex based or human managed redaction processes.

The multi format input support (TXT, DOCX, PDF) of the system, with three standalone anonymization modes (highlight, replace, and redact), is unrivaled for flexibility within privacy preservation operations. It comes with a self explanatory Streamlit GUI so that the tool remains within the reach of non technical as well as technical users. Further, the deployment also ensures that important post processing parameters such as readability, semantic retention, and coverage are quantitatively assessed using measures such as Flesch-Kincaid Readability Score, Lexical Density, and an in house Semantic Retention Index.

Empirical findings show as much as 30% accuracy improvement in NER, 82% semantic preservation improvement, and more than 20% redaction coverage boost compared to traditional approaches. Such findings illustrate how the proposed system delivers a tidy, accurate, and easy to use solution to maintaining privacy, particularly in environments where compliance, document readability, and business efficiency are critical.

REFERENCES

[1] Revathy, G., Nandhini, T., Senthilvadivu, S., & Mariyammal, M. (2024, December). Integrating Federated Learning and IoT for Privacy-Preserving Smart Healthcare Systems. In *2024 International Conference on Sustainable Communication Networks and Application (ICSCNA)* (pp. 141-145). IEEE.

[2] Biswas, Sreemoyee, et al. "Machine learning concepts for correlated Big Data privacy." *Journal of Big Data* 8.1 (2021): 157.

[3] United Nations: A world that counts. Mobilizing the data revolution for sustainable development. United Nations, New York (2014)

[4] Rane, Prachi, et al. "Redacting sensitive information from the data." *2021 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)*. IEEE, 2021.

[5] https://rivery.io/blog/big-data-statistics-how-much-data-is-there-in-the-world/

[6] Shanmugapriya, S. S., Anila, S. V., Hemalatha, D., Karthika, K., Sankaradass, V., & Singh, S. (2024, April). Optimizing Security Posture via Synergized Fingerprint and Facial Recognition in Integrated AuthenticationFramework. In *2024 International Conference on Advances in Data Engineering and Intelligent Computing Systems (ADICS)* (pp. 1-5). IEEE.

[7] Zhu, F., Yi, X., Abuadbba, A., Khalil, I., Nepal, S., & Huang, X. (2021). Cost-effective authenticated data redaction with privacy protection in IoT. *IEEE Internet of Things Journal*, *8*(14), 11678-11689.

[8] Peng, S., Huang, M. J., Wu, M., & Wei, J. (2024). Transforming Redaction: How AI is Revolutionizing Data Protection. *arXiv preprint arXiv:2409.15308*.

[9] Gusain, V., & Leith, D. (2025). Improving Privacy Benefits of Redaction. *arXiv preprint arXiv:2501.17762*.

[10] Zhou, L., Yu, L., Zou, J., & Min, H. (2023, July). Privacy-Preserving Redaction of Diagnosis Data through Source Code Analysis. In *Proceedings of the 35th International Conference on Scientific and Statistical Database Management* (pp. 1-4).

[11] Lopez, A. B. (2022). Adult Guardianship Privacy, Redaction, and Professional Responsibility. *ACTEC LJ*, *48*, 77.

[12] Thetbanthad, P., Sathanarugsawait, B., & Praneetpolgrang, P. (2025). Automated Redaction of Personally Identifiable Information on Drug Labels Using Optical Character Recognition and Large Language Models for Compliance with Thailand's Personal Data Protection Act. *Applied Sciences*, *15*(9), 4923.

[13] Andersson, J., & Aguilar Aguilar, C. (2022). Attribute-Based Content Redaction in Large-Scale Data Systems: A Study of Granular Access Control.

[14] Determann, L. (2019). Privacy and Data Protection. *Московский журнал международного права*, (1), 18-26.